Centripetal Force and Gravitation: Example

SYNOPSIS

An optional in-class assignment, and a single question: **confirm that the Moon's orbital period is about a month.**

Nothing new is needed. Gravity is the only force on the Moon, its orbit is very nearly circular, and a circular orbit is a body in uniform circular motion — so the gravitational force *is* the centripetal force. Set the two expressions equal, cancel what cancels, and the orbital speed drops out. Divide the circumference by it and you have the period.

The thing worth watching is what cancels. The Moon's own mass appears on both sides and goes, which is why the answer does not depend on it — and that is the same cancellation that makes every satellite at a given radius orbit at the same speed, whatever it weighs.

BEFORE YOU START — PREDICT

Two satellites orbit the Earth at the same altitude, one ten times as massive as the other. Write one sentence saying which of them takes longer to go round.

PERFORMANCE EXPECTATIONS

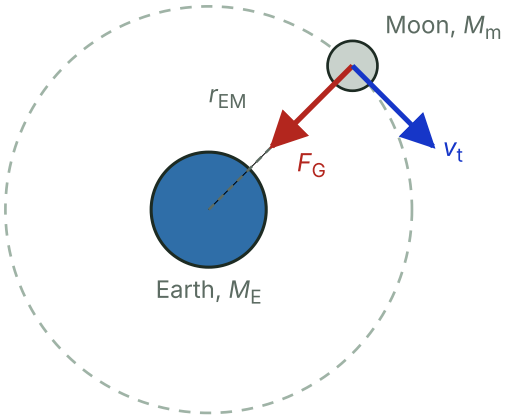
Recognise a circular orbit as uniform circular motion with gravity supplying the centripetal force. Set $F_G = F_c$ and solve for the orbital speed. Convert an orbital speed to an orbital period. State how the period depends on the central mass and on the orbital radius.

WATCH · THE SHEET WORKED THROUGH

Recording to come. Print the sheet (link at the bottom of this page), make your prediction, then work the items as the video goes through them.

Set the two forces equal

The Moon goes round the Earth on a very nearly circular path. One force acts on it, and it points at the Earth — which is exactly where a centripetal force has to point. So the two are the same force called two names, and that is the whole method.



a circular orbit is uniform circular motion

one force, two names

$$GM_EM_m/r^2 \quad = \quad M_mv_t^2/r$$

M_m cancels; one power of r cancels

$$v_t = \sqrt{(GM_E/r_{EM})}$$

no M_m anywhere in the answer

The Moon's own mass is on both sides of the first line and gone by the second. That cancellation is the reason a heavy satellite and a light one at the same radius keep exactly the same company.

Here is what you are given, and what you are after.

Quantity	Value
gravitational constant G	6.674×10^{-11} N·m²/kg²
mass of the Earth M_E	5.972×10^{24} kg
mass of the Moon M_m	7.348×10^{22} kg
Earth–Moon distance r_{EM}	384 400 km = 3.844×10^{8} m
orbital speed v_t	
orbital period T	

1. Write the gravitational force on the Moon and the centripetal force on the Moon, set them equal, and cancel everything that cancels. What is left?
2. One of the four given quantities turns out not to be needed. Which one, and why is that worth noticing rather than annoying?

3. Work out the orbital speed v_t, in m/s, and fill it into the table.

Show the working

From the speed to the period

One orbit is one circumference at that speed. That is the entire second half, and the only thing that can go wrong is the arithmetic of converting seconds into something you can recognise.

4. How far does the Moon travel in one orbit, in metres? Then find the period T, in seconds, and fill it into the table.
5. Convert that to days. Is it about a month?
6. **For the brownie point.** Without putting any numbers in, combine your two results into a single expression for T in terms of G, M_E and r. How does the period depend on the mass of the Earth, and how does it depend on the orbital radius?

Show the working

Looking ahead

The same two lines answer a surprising number of questions. Run them backwards and the Moon's orbit *weighs the Earth*; put in T = 24 h and they give you the altitude a communications satellite has to sit at; swap the Earth for the Sun and they give you the Earth's year. **Note 3** leaves orbits behind and turns to bodies that spin rather than circle.

BACK TO YOUR PREDICTION

Neither — they take exactly the same time. The orbiting body's mass is on both sides of $GM_EM/r^2 = Mv^2/r$ and cancels, so nothing about the satellite itself survives into the answer. Only the Earth's mass and the orbital radius do. If you reasoned “heavier means more gravitational pull, so faster”, you were half right: the pull *is* ten times bigger, and so is the inertia it has to overcome.

Check yourself

1. Using $GM_E = 3.986 \times 10^{14}$ and $r = 3.844 \times 10^8$ m, what is the Moon's orbital speed, in m/s?

_____ m/s

2. The orbital speed of a satellite in a circular orbit depends on

- ○ the satellite's mass and the orbital radius
- ○ the central body's mass and the orbital radius
- ○ both masses and the orbital radius
- ○ both masses only

3. A satellite is moved to a circular orbit of twice the radius. Its period changes by a factor of

- ○ 2
- ○ $\sqrt{2} \approx 1.41$
- ○ $2^{3/2} \approx 2.83$
- ○ 4

4. At 1018 m/s round a circumference of 2.415×10^9 m, how long is one orbit, in days?

_____ days

Check Clear

Your notes

IN CLASS

The sheet — blank (PDF)

print or open on a tablet; make your prediction, then work the items.

AFTER THE MODULE

Completed sheet (PDF)

released when Module 5 closes.

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